

- What design do I have space for? For example, do other components on the board dictate this?

- What is the end product that the circuit is required for?

- Which regulations need to be followed—industrial, railway or medical?

- In which environmental conditions will the application be used?

- How reliable should or must the whole circuit and application be?

So, while a Porsche 911 is a great car, it is easy to understand that it would not be suitable for an expedition through the South American rainforest, where a four-wheel drive jeep would probably be the better option.

But selecting the appropriate power supply is not always as simple as choosing the right car. Once the standard basic questions have been cleared up, we come to the ones that this article intends to investigate more fully:

- How can I measure correctly and avoid errors?

- How do I deal with ripple and noise?

- What happens in the event of inrush current?

- What do I need to be aware of with respect to electromagnetic compatibility (EMC)?

As an example, Traco Power's product portfolio includes more than

25 3W DC/DC converters, such as the TVN series with ultra-low ripple and noise, the THM series (certified for medical applications and open-frame variants) or the TMR-WIR series (approved for railway applications with 3000V DC isolation voltage) open up a whole host of possibilities.

We would probably all like a converter that meets all necessary requirements in one go—an off-the-shelf solution, but usually we need to spend more. This is because we need more than one converter to meet the requirements or additional wiring to achieve the values we require.

So, what do we need to establish to find a solution? The first step is to determine which measurement values we need. A simple measurement allows us to easily make a rough assessment—input and output voltage and current on the converter, on the load and possible changes in efficiency.

Avoiding measurement errors

A basic point to remember is that every measurement changes the actual state of the circuit, and any impact needs to be kept as minimal as possible. This means that doing a four-wire measurement even for a simple measurement is advisable. Measuring current and voltage with independent test leads means that the leads' inherent resistance will have less impact on values.

It is also important to think about end usage. For example, in an operating theatre, there can be 30m wires between the power supply and the actual load. If the load required is 24V, the source needs a higher output voltage to offset voltage loss in the wire. That means that both load and source need to be

measured. Fig. 1 shows an example of a classic four-wire voltage measurement at source.

Establishing and influencing ripple and noise

Why are ripple and noise measured. Regardless of application, ripple and noise of a DC/DC converter could lie in its actual area of operation, for example, in a measuring bridge. As a result, this needs to be considered and evaluated separately.

What are these, and how can these be measured cleanly. We talk about ripple in AC/DC and DC/DC circuits when irregular disturbances are caused by internal circuits. Whereas, noise denotes the peaks that return periodically, produced by the transforming pulsing at switching frequency.

To determine actual values, the probe head must be in direct contact with the pins, the ground ring and the measurement tip, making contact with these. To be able to compare results with a manufacturer's data, bandwidth on the oscilloscope is limited to 20MHz, a common value for laboratory work (Fig. 2).

Usually, noise and ripple can simply be reduced with two parallel-switched capacitors, for example, a 100nF metal-film capacitor and a 10µF electrolytic capacitor. This is done while always bearing in mind that values presented on datasheets can be influenced by other factors during end usage.

Dealing with inrush current

This information is important for ensuring that components upstream are of the correct dimensions.

- Current depends essentially on switching speed so, ideally, mercury switches should be used in the laboratory.

- Source should have lowest internal resistance possible.

- Current is measured with a demagnetised tip.

Ambient temperature also has a big impact on inrush current. For example, use of electrolytic capacitors is highly dependent on temperature.

Fig. 3 shows an example of inrush voltage in an LED lamp (yellow line). It also shows the voltage pattern in

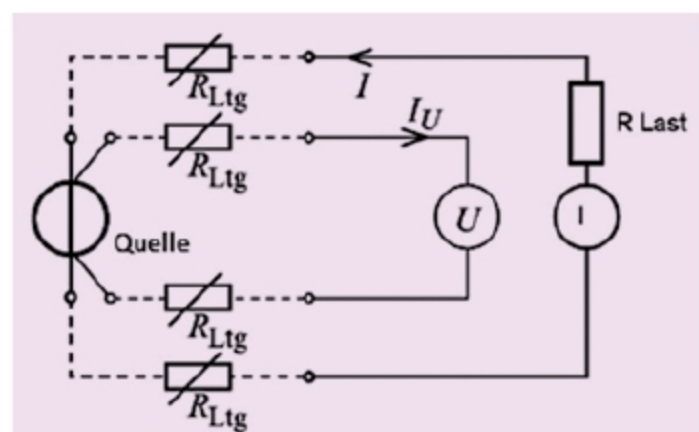


Fig. 1: Circuit diagram of a four-wire measurement with source and load

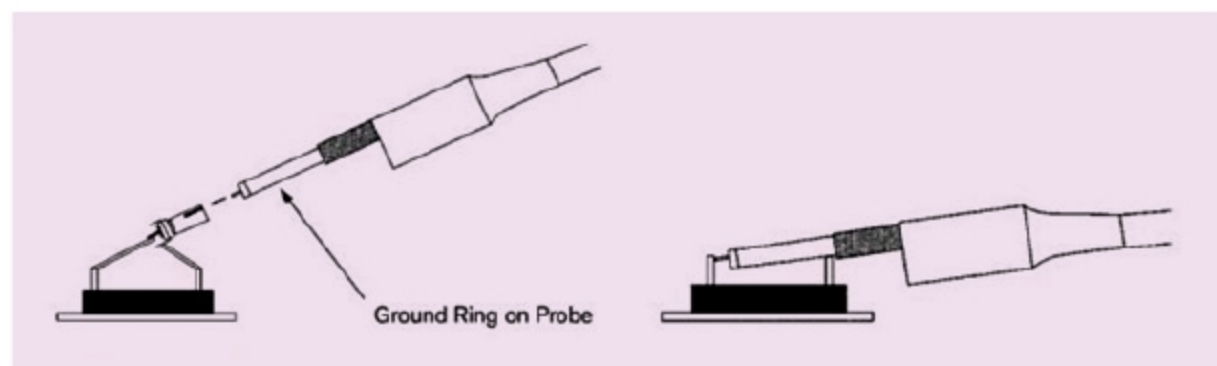


Fig. 2: Correct application of measurement tips to a DC/DC converter with pins